

Optimization Methods in Machine Learning, CUB, Spring 2026

Theory 2. Convex sets and functions.

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March 17, 2026

1. Task 1

Let $z = \lambda x + (1 - \lambda)y$, $\lambda \in [0, 1]$.

For $\text{cl}(C)$: take $x_k, y_k \in C$ such that

$$x_k \rightarrow x, \quad y_k \rightarrow y.$$

Since C is convex,

$$\lambda x_k + (1 - \lambda)y_k \in C.$$

Passing to the limit,

$$\lambda x_k + (1 - \lambda)y_k \rightarrow z,$$

hence $z \in \text{cl}(C)$.

For $\text{int}(C)$: if $\lambda = 0$ or 1 , trivial. Let $0 < \lambda < 1$. Take open sets U, V such that

$$x \in U \subseteq C, \quad y \in V \subseteq C.$$

Then $\lambda U + (1 - \lambda)V$ is open, contains z , and by convexity of C ,

$$\lambda U + (1 - \lambda)V \subseteq C.$$

So $z \in \text{int}(C)$.

Therefore $\text{cl}(C)$ and $\text{int}(C)$ are convex.

2. Task 2

2.1. (a)

Since $C \subseteq \text{Conv}(C)$ and $D \subseteq \text{Conv}(D)$,

$$\text{Conv}(C \cup D) \subseteq \text{Conv}(\text{Conv}(C) \cup \text{Conv}(D)).$$

Also $\text{Conv}(C \cup D)$ is convex and contains both C and D , hence it contains $\text{Conv}(C)$ and $\text{Conv}(D)$.
So

$$\text{Conv}(\text{Conv}(C) \cup \text{Conv}(D)) \subseteq \text{Conv}(C \cup D).$$

Therefore

$$\text{Conv}(C \cup D) = \text{Conv}(\text{Conv}(C) \cup \text{Conv}(D)).$$

2.2. (b)

Since $C \cap D \subseteq C$ and $C \cap D \subseteq D$,

$$\text{Conv}(C \cap D) \subseteq \text{Conv}(C), \quad \text{Conv}(C \cap D) \subseteq \text{Conv}(D).$$

Hence

$$\text{Conv}(C \cap D) \subseteq \text{Conv}(C) \cap \text{Conv}(D).$$

3. Task 3

Let

$$S = \left\{ x \in \mathbb{R}^n : x^T P x \leq (c^T x)^2, c^T x \geq 0 \right\}.$$

Since $P \succ 0$,

$$x^T P x = \|P^{\frac{1}{2}} x\|_2^2.$$

Because $c^T x \geq 0$,

$$x \in S \iff \|P^{\frac{1}{2}} x\|_2 \leq c^T x.$$

Take $x, y \in S, \lambda \in [0, 1]$. Then

$$\|P^{\frac{1}{2}}(\lambda x + (1 - \lambda)y)\|_2 \leq \lambda \|P^{\frac{1}{2}} x\|_2 + (1 - \lambda) \|P^{\frac{1}{2}} y\|_2 \leq \lambda c^T x + (1 - \lambda)c^T y = c^T(\lambda x + (1 - \lambda)y).$$

Hence $\lambda x + (1 - \lambda)y \in S$, so S is convex.

4. Task 4

4.1. (a)

Write

$$S = S_1 \cap S_2 \cap S_3,$$

where

$$S_1 = \{X \in S^n : A^T X + X A + I \preceq 0\},$$

$$S_2 = \{X \in S^n : \text{tr}(X) \leq 1\},$$

$$S_3 = \{X \in S^n : X \succeq 0\}.$$

S_1 is convex because $A^T X + X A + I$ is affine in X . S_2 is a halfspace. S_3 is the PSD cone. Therefore S is convex.

4.2. (b)

Since $X \succeq I$, X is positive definite. By Schur complement,

$$u^T X^{-1} u \leq 1 \iff \begin{pmatrix} X & u \\ u^T & 1 \end{pmatrix} \succeq 0.$$

So the set equals

$$\left\{ (u, X) : \begin{pmatrix} X & u \\ u^T & 1 \end{pmatrix} \succeq 0, I \preceq X \preceq 2I \right\}.$$

This is an intersection of LMIs, hence convex.

5. Task 5

Let

$$K = \{xx^T : \|x\|_2 = 1\}.$$

If

$$A = \sum_{i=1}^m \theta_i x_i x_i^T, \quad \theta_i \geq 0, \quad \sum_{i=1}^m \theta_i = 1,$$

then

$$A \succeq 0,$$

and

$$\text{tr}(A) = \sum_{i=1}^m \theta_i \text{tr}(x_i x_i^T) = \sum_{i=1}^m \theta_i \|x_i\|_2^2 = \sum_{i=1}^m \theta_i = 1.$$

Hence

$$\text{Conv}(K) \subseteq \{A \in S_+^n : \text{tr}(A) = 1\}.$$

Conversely, let $A \in S_+^n$ and $\text{tr}(A) = 1$. Spectral decomposition gives

$$A = \sum_{i=1}^n \lambda_i q_i q_i^T,$$

where

$$\lambda_i \geq 0, \quad \|q_i\|_2 = 1, \quad \sum_{i=1}^n \lambda_i = \text{tr}(A) = 1.$$

So A is a convex combination of matrices $q_i q_i^T \in K$. Therefore

$$\text{Conv}(K) = \{A \in S_+^n : \text{tr}(A) = 1\}.$$

6. Task 6

=> If f is convex, then for $t \in (0, 1]$,

$$f(x + t(y - x)) \leq (1 - t)f(x) + tf(y).$$

Hence

$$\frac{f(x + t(y - x)) - f(x)}{t} \leq f(y) - f(x).$$

Let $t \rightarrow 0 +$:

$$\nabla f(x)^T (y - x) \leq f(y) - f(x).$$

So

$$f(y) \geq f(x) + \nabla f(x)^T (y - x).$$

<= Assume

$$f(y) \geq f(x) + \nabla f(x)^T (y - x), \quad \forall x, y.$$

Take

$$z = (1 - \lambda)x + \lambda y.$$

Then

$$f(x) \geq f(z) + \nabla f(z)^T(x - z),$$

$$f(y) \geq f(z) + \nabla f(z)^T(y - z).$$

Multiply by $1 - \lambda$ and λ and add:

$$(1 - \lambda)f(x) + \lambda f(y) \geq f(z) + \nabla f(z)^T((1 - \lambda)(x - z) + \lambda(y - z)) = f(z).$$

Hence

$$f((1 - \lambda)x + \lambda y) \leq (1 - \lambda)f(x) + \lambda f(y),$$

so f is convex.

7. Task 7

7.1. (a)

Let

$$g_{i(t)} = \max(0, t)^2.$$

The map $t \mapsto \max(0, t)$ is convex and nondecreasing, and $s \mapsto s^2$ is convex and nondecreasing on $\mathbb{R}_+$. Hence each g_i is convex.

Now define

$$L(z) = \log \left(\sum_{i=1}^n \exp(z_i) \right).$$

L is convex and nondecreasing in each coordinate. Therefore

$$f(x) = L(g_1(x_1), \dots, g_n(x_n))$$

is convex.

7.2. (b)

Let

$$M = \{P \in S^n : I \preceq P \preceq 5I\}.$$

Then

$$\text{epi } f = \{(x, t) : \exists P \in M, x^T P^{-1} x + \text{tr}(P) \leq t\}.$$

Since $P \succeq I$, P is positive definite, and by Schur complement

$$x^T P^{-1} x + \text{tr}(P) \leq t \iff \begin{pmatrix} P & x \\ x^T & t - \text{tr}(P) \end{pmatrix} \succeq 0.$$

Hence

$$\text{epi } f = \text{proj}_{x,t} \left\{ (x, t, P) : P \in M, \begin{pmatrix} P & x \\ x^T & t - \text{tr}(P) \end{pmatrix} \succeq 0 \right\}.$$

The set inside is convex (LMIs + affine terms), and projection preserves convexity. So $\text{epi } f$ is convex, hence f is convex.

8. Task 8

For

$$f(x) = \prod_{i=1}^n x_i^{\alpha_i}, \quad \alpha_i \geq 0, \quad \sum_{i=1}^n \alpha_i = 1,$$

set

$$a_i = \frac{\alpha_i}{x_i}.$$

Then

$$\nabla f(x) = f(x)(a_1, \dots, a_n)^T,$$

$$\nabla^2 f(x) = f(x) \left(aa^T - \text{diag} \left(\frac{\alpha_1}{x_1^2}, \dots, \frac{\alpha_n}{x_n^2} \right) \right).$$

So for any $h \in \mathbb{R}^n$,

$$h^T \nabla^2 f(x) h = f(x) \left[\left(\sum_{i=1}^n \alpha_i \frac{h_i}{x_i} \right)^2 - \sum_{i=1}^n \alpha_i \left(\frac{h_i}{x_i} \right)^2 \right].$$

Let $z_i = \frac{h_i}{x_i}$. Then

$$\left(\sum_{i=1}^n \alpha_i z_i \right)^2 \leq \left(\sum_{i=1}^n \alpha_i \right) \left(\sum_{i=1}^n \alpha_i z_i^2 \right) = \sum_{i=1}^n \alpha_i z_i^2.$$

Hence

$$h^T \nabla^2 f(x) h \leq 0.$$

So $\nabla^2 f(x) \preceq 0$, therefore f is concave.

9. Task 9

Define

$$g_{n(x)} = x_n,$$

$$g_{k(x)} = x_k - \frac{1}{g_{k+1}}(x), \quad k = n-1, \dots, 1.$$

Then

$$f(x) = \frac{1}{g_1}(x).$$

Also define

$$D_n = \{x : x_n > 0\},$$

$$D_k = \{x \in D_{k+1} : g_{k(x)} > 0\}.$$

So the domain is D_1 .

Now induct on k : g_n is affine, hence concave. If g_{k+1} is concave and positive on D_{k+1} , then $\frac{1}{g_{k+1}}$ is convex on D_{k+1} because $t \mapsto \frac{1}{t}$ is convex and decreasing on $\mathbb{R}_{++}$. Therefore

$$g_k = x_k - \frac{1}{g_{k+1}}$$

is concave on D_{k+1} , and

$$D_k = \{x \in D_{k+1} : g_{k(x)} > 0\}$$

is convex.

So g_1 is positive and concave on the convex set D_1 . Again $t \mapsto \frac{1}{t}$ is convex and decreasing on $\mathbb{R}_{++}$, hence

$$f = \frac{1}{g_1}$$

is convex on D_1 .

10. Task 10

=> If f is convex, then with $\lambda = \frac{1}{2}$,

$$f\left(\frac{x+y}{2}\right) \leq \frac{f(x) + f(y)}{2}.$$

<= Assume

$$f\left(\frac{u+v}{2}\right) \leq \frac{f(u) + f(v)}{2}, \quad \forall u, v \in \mathbb{R}^n.$$

For every dyadic $\lambda = \frac{m}{2^k}$, repeated use of the midpoint inequality gives

$$f((1-\lambda)x + \lambda y) \leq (1-\lambda)f(x) + \lambda f(y).$$

Now take arbitrary $\lambda \in [0, 1]$ and choose dyadics $\lambda_j \rightarrow \lambda$. By continuity,

$$f((1-\lambda)x + \lambda y) = \lim_{j \rightarrow \infty} f((1-\lambda_j)x + \lambda_j y) \leq \lim_{j \rightarrow \infty} ((1-\lambda_j)f(x) + \lambda_j f(y)) = (1-\lambda)f(x) + \lambda f(y).$$

Hence f is convex.